

Diagnosing LLM Errors with Controlled Evidence Interventions: A Benchmark and Arithmetic-Alignment Method

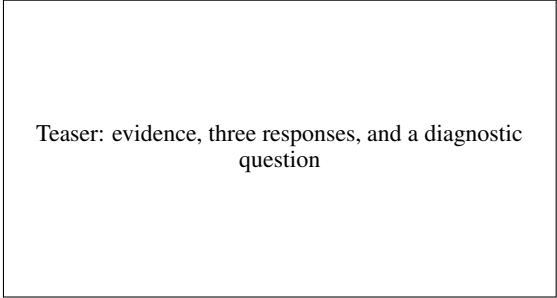
Discussion draft

Abstract

An answer can be confident, repeatable, and wrong. In document-grounded question answering, controlled evidence changes offer a way to probe such errors. Yet even a response that moves in the expected direction may follow the wrong calculation. The challenge is to connect response behavior to the evidence that could explain it. We introduce a reusable financial question answering benchmark linking dependency-aware numerical edits, saved three-world responses, and generation-aligned error labels. It contains 1,699 human-reviewed constructions and supports both offline detector evaluation and reconstruction of model inputs. Building on this benchmark, Program-Executable Counterfactual Reliability (PECR) augments bidirectional responses with arithmetic alignment: which cell-level calculations match each answer, whether those matches persist across interventions, and how they relate to the question. A compact 44-dimensional representation requires no additional model calls. In grouped evaluations on Qwen3.5-4B and DeepSeek V4.1 Flash, PECR improves AU-ROC over a strong three-response baseline by 2.59 and 2.50 points, with positive adjusted paired intervals. Construction-information controls and component ablations locate the additional signal beyond edit magnitude and response geometry. The benchmark and method make evidence-dependent error diagnosis reproducible and open a path toward more informative verification under limited query budgets.

1 Introduction

Confidence and agreement can make an incorrect answer look reliable. For document-grounded question answering, this is consequential: a plausible number may inform an analyst’s decisions before the underlying calculation is checked.



Teaser: evidence, three responses, and a diagnostic question

Figure 1: Does an answer respond for the right numerical reason? Opposite evidence edits reveal how an answer changes; arithmetic correspondences additionally describe which calculations remain compatible with those responses. A matching calculation is diagnostic evidence, not a certificate of correctness.

Sampling-based consistency, semantic uncertainty, and confidence estimation provide useful signals for identifying unreliable outputs (Manakul et al., 2023; Farquhar et al., 2024; Geng et al., 2024). Nevertheless, models can repeatedly produce the same incorrect answer, leaving self-consistent errors difficult to detect (Tan et al., 2025). This motivates a complementary question: can the way an answer responds to changes in its supporting evidence reveal an error that confidence and agreement fail to expose?

Recent benchmarks provide systematic settings for evaluating uncertainty estimators, predicting model errors, and testing selective refusal under flawed context (Vashurin et al., 2025; Pacchiardi et al., 2025; Muhamed et al., 2026). We focus on a more specific connection: whether responses to controlled evidence changes help diagnose the correctness of the original answer. Studying this connection requires more than a collection of questions and reference answers. It requires specified interventions, their response trajectories, and correctness labels tied to the original generations, so that detectors can be compared using the same evidence and response budget.

Numerical question answering makes this connection concrete. Perturbation-based uncertainty estimation and numerical veracity probes demonstrate the value of testing models under modified inputs (Gao et al., 2024; Aarnes and Setty, 2025). A meaningful numerical edit, however, often calls for a predictable change rather than invariance: increasing a numerator and increasing a denominator imply opposite response directions. Even the expected direction is insufficient. A model may consistently use the wrong denominator, or calculate from a row belonging to another year. Its answers can move smoothly across interventions while the original answer remains wrong.

Our central observation is that a response trajectory becomes more informative when paired with its *arithmetic correspondence to the evidence*. Instead of only asking how much the answer changes, we ask which simple calculations could produce it, whether the same calculation remains compatible in all three worlds, and whether the associated rows, years, and units fit the question. This makes a distinction that scalar response geometry cannot make on its own: similar trajectories can correspond to different evidence relationships. Figure 1 illustrates this diagnostic question.

We introduce a controlled-intervention benchmark together with PECR, an error detector that represents these relationships. The benchmark turns FinQA examples into linked records of numerical evidence edits, model responses, and original-answer labels. Its construction process updates evidence-determined dependencies, including totals and repeated mentions, rather than changing an isolated operand alone. All 1,699 materialized constructions received a recorded human validity judgment. Versioned edit patches, response bindings, complete coverage records, and fixed group splits support both comparisons on saved responses and reconstruction of inputs for another model.

PECR combines the three-world response representation with *Arithmetic44*: per-world numerical alignment, cross-world persistence of candidate calculations, and question-operation cues. The extractor enumerates bounded one-step arithmetic candidates from the supplied tables; it neither reads reference answers nor executes the annotated solution program. A fixed histogram gradient boosting (HGB) predictor learns which combinations indicate original-answer errors. This adds evidence correspondence at the same three-

response budget, without an auxiliary verifier or access to hidden states.

Contributions. (1) We provide a reusable benchmark for original-answer error diagnosis with human-reviewed, dependency-aware interventions, generation-aligned targets, and verifiable input reconstruction. (2) We introduce an arithmetic-alignment representation that connects three-world behavior to cell-level calculations and question context. (3) Matched-budget evaluations yield AUROC gains of 2.59 points on Qwen3.5-4B and 2.50 on DeepSeek V4.1 Flash over a strong Raw comparator. New construction-information controls separate the value of edited responses from edit descriptors; ablations reveal different useful feature blocks across the two models. Together, these results support evidence correspondence as a productive direction beyond response geometry alone.

2 Related Work

Uncertainty and error prediction. SelfCheck-GPT estimates factual reliability from agreement across sampled generations (Manakul et al., 2023); semantic entropy groups generations by meaning (Farquhar et al., 2024). Black-box confidence methods combine additional response signals (Chen and Mueller, 2024), while systematic studies examine estimator reliability and robustness (Mahaut et al., 2024; Vashurin et al., 2025). PredictaBoard evaluates assessors that anticipate model failures (Pacchiardi et al., 2025). Self-consistent errors motivate using information beyond repeated-answer agreement, including cross-model hidden-state probes (Tan et al., 2025). PECR instead uses controlled input edits and deterministic evidence-answer correspondences. Our comparisons isolate that representation under a fixed response budget.

Perturbations and behavioral evaluation. CheckList and contrast sets test expected behavior beyond aggregate accuracy (Ribeiro et al., 2020; Gardner et al., 2020). SPUQ varies inputs and sampling conditions for uncertainty estimation (Gao et al., 2024). GSM-Symbolic and NumPert expose weaknesses under numerical changes (Mirzadeh et al., 2025; Aarnes and Setty, 2025), while RefusalBench uses controlled perturbations to assess selective refusal (Muhammed et al., 2026). We connect such behavioral evidence to a distinct

target: the error of a particular original generation. Dependency-aware construction matters because an internally inconsistent intervention can make an apparently anomalous answer reasonable.

Numerical reasoning resources. FinQA supplies expert questions and executable reasoning programs over financial reports (Chen et al., 2021); MultiHiertt extends numerical reasoning to hierarchical tables and text (Zhao et al., 2022). We build on existing financial questions rather than collect a new document corpus. The added resource is the binding among interventions, response trajectories, model-specific error targets, and reproducible detector evaluation. Arithmetic44 uses simple candidate calculations as features, with no requirement that a candidate reconstruct the annotated solution.

3 The Controlled-Intervention Benchmark

3.1 Task and response-specific targets

An item contains a question and evidence $x_i = (q_i, E_i)$ with source group g_i . For answer model m , we save

$$r_{im}^w = (a_{im}^w, c_{im}^w, u_{im}^w), \quad w \in \{-, 0, +\}, \quad (1)$$

where a , c , and u denote numerical answer, reported confidence, and unit. Each world is a separate call. The detector ranks errors in a_{im}^0 , using the original and the two edited responses. It does not predict whether the edited answers are correct.

Labels are tied to the saved original generation. If the original answer and FinQA reference a_i^* are comparable under the frozen numerical parser, then

$$y_{im} = \mathbb{I}[|a_{im}^0 - a_i^*| > \max(10^{-4}, 10^{-4}|a_i^*|)]. \quad (2)$$

Otherwise the label is unknown. The parser removes supported surface markers without percentage rescaling or unit conversion; this is free-text numerical scoring, not official FinQA program-execution accuracy. Reference answers are accessed only in the separate labeling stage. Response hashes bind labels, original-side features, and all three endpoints to the same generation. This prevents a detector from learning the error of an earlier answer using the features of a later one.

Table 1: Model-specific coverage in the repaired benchmark.

Population	Qwen	DeepSeek
Attempted	1,699	1,597
Strict	1,597	1,537
Source groups (strict)	413	407
Errors (strict)	1,258	837
Correct (strict)	339	700
Attempted, excluded	102	60
Strict / attempted	94.0%	96.2%

3.2 Dependency-aware three-world construction

The starting frame contains 2,241 previously explored FinQA TRAIN items; 2,225 were requested for construction and 16 were not. A program-to-evidence mapping identifies a seed operand b_i and supplies an expected forward direction $e_i \in \{-1, +1\}$. The two interventions change the seed in opposite directions. They preserve the original question and update dependent evidence where needed.

Construction proceeds in two stages. A drafting model proposes complete edits to both worlds; unresolved drafts receive a second model pass. The constructors see the source excerpt and saved edit specification, without target-model responses, error labels, or reference answers. They must preserve units, periods, table structure, and the original world; propagate evidence-supported changes through totals, roll-forwards, ratios, and repeated mentions; and flag dependencies that cannot be resolved from the excerpt. A deterministic compiler applies the edits against exact source values and records the resulting world hashes. Appendix B gives the construction and review provenance.

This procedure yields 1,699 complete drafts, with 525 requiring further completion and one unconstructable. The retained inputs include 1,126 items with multiple table-cell edits in at least one world and 389 with textual updates. Thus, most interventions involve more than replacing a single number. One human reviewer assessed all 1,699 complete drafts as valid. Separately, 44 retained items require direction rederivation and are excluded from the strict detector evaluations. Validity judgments concern the constructed evidence; they do not depend on whether a detector improves.

3.3 Model tracks and coverage

The Qwen3.5-4B track attempted all 1,699 complete constructions. After direction, parsing, feature, and label requirements, 1,597 items in 413 source groups form its strict cohort. The DeepSeek V4.1 Flash track attempted those 1,597 Qwen-strict items; 1,537 in 407 groups meet its own strict requirements (Table 1). All representations within a track use exactly the same items. Candidate arithmetic matches are not an eligibility condition.

The two tracks have different targets: 449 of their 1,537 shared strict items receive different error labels. They therefore constitute separate detector evaluations. DeepSeek inherits the common items’ saved fold assignments, while its responses, labels, and original-side features are native to DeepSeek. Neither labels nor original-answer features are borrowed across models.

3.4 Two forms of reuse

The release supports both *offline risk prediction* and *new-model intervention evaluation*. The former uses numerical matrices, separate labels, saved folds, and reference out-of-fold (OOF) scores. The latter rebuilds the three-world requests from versioned patches, prompt profiles, and the exact FinQA source. A local reconstruction audit matches all 9,888 saved request hashes: 5,097 Qwen and 4,791 DeepSeek world slots. A user can submit these requests through an external client, then generate labels and features bound to the new answers. This preserves the intervention while allowing the response population to change.

The broader frame remains available for coverage accounting (Appendix B). Unknown parsing and label states are not converted to successes. The artifact separates immutable source records from derived tables and provides positional feature schemas, source hashes, and an offline retraining entry point. These bindings allow a changed result to be traced to its inputs, generation, features, or learner rather than treated as an unexplained headline difference.

4 PECR: Arithmetic Alignment Across Worlds

The method asks whether a response trajectory is compatible with the same arithmetic evidence across interventions, and whether that evidence fits the question. PECR retains the raw response

signal and adds Arithmetic44, a fixed representation of candidate calculations. Figure 3 summarizes the computation. We suppress model and item subscripts below.

4.1 A strong response-based foundation

The raw response block encodes forward changes

$$v^+ = a^+ - a^0, \quad v^- = a^0 - a^-, \quad (3)$$

their magnitudes and signed relations to e , flatness, symmetry, confidence changes, and unit agreement. Together with the direction flag, these form R_{16} . We also supply the actual seed edit and its relative scale,

$$h = b^+ - b^0, \quad t = \frac{h}{|b^0| + \varepsilon}. \quad (4)$$

The original-side vector $G \in \mathbb{R}^{96}$ contains typed numerical, structural, lexical, and answer-derived summaries. Our matched Raw baseline is

$$B = \text{concat}(G, R_{16}, h, t) \in \mathbb{R}^{114}. \quad (5)$$

Thus, the baseline already sees the same three responses, expected direction, and edit scale. Arithmetic44 must add value beyond these inputs, not merely recover information withheld from the comparator. Appendix A specifies the columns, including the historical Raw17 compatibility distinction.

4.2 Candidate calculations as evidence correspondences

Let $\mathcal{V}$ contain table coordinates with finite numerical values in all three worlds, retaining at most the first 128 in row-major order. Headers are excluded. A candidate $\pi = (o, j, k)$ combines distinct cells $j, k \in \mathcal{V}$ using an allowed operation o . The core operations are sum, difference, and ratio. When all three reported answer units indicate percentages, ratio is expressed on a percentage scale and percentage change is additionally allowed. Candidates with zero denominators or nonfinite results in any world are discarded; duplicate sums are removed.

Write $\pi(E^w)$ for the candidate’s result in world w . Its normalized discrepancy from the saved answer is

$$d_w(\pi) = \frac{|\pi(E^w) - a^w|}{\tau_w}, \quad (6)$$

$$\tau_w = \max(0.01, 10^{-4}|a^w|).$$

Benchmark: reviewed constructions, model-specific responses, aligned targets, and reusable evaluation

Figure 2: Benchmark organization. Complete edit specifications reconstruct the original and two edited model requests. Each model track binds its own responses, labels, features, and folds. Coverage records retain excluded attempts as well as unattempted construction candidates.

PECR: three-world probing $\rightarrow$ arithmetic candidates and alignment $\rightarrow$ feature fusion $\rightarrow$ original-answer risk

Figure 3: PECR augments Raw with 44 arithmetic-alignment features: 3×8 per-world features, 16 cross-world features, and four question-operation cues. Gold-derived error labels supervise the training folds only. All feature extraction uses the supplied questions, worlds, responses, and edit location.

The world-specific match set is

$$\mathcal{M}^w = \{\pi \in \mathcal{P} : d_w(\pi) \leq 1\}, \quad (7)$$

where $\mathcal{P}$ is the bounded candidate pool. The same tuple identifies a candidate across worlds. Matching tolerance in Equation 6 defines a feature, whereas Equation 2 defines the separately computed target; the two serve different purposes.

The extractor does not select a single explanation. It preserves ambiguity through match counts, uniqueness, and set relations. An absent match is likewise represented rather than filtered out. Numerical compatibility need not mean the model actually used those cells, or that the calculation answers the question. The following features make those distinctions available to the learner.

4.3 Worldwise alignment and question context

For each world, an eight-dimensional map ψ_w summarizes $\mathcal{M}^w$: log match count; uniqueness; the fraction involving the seed operand; maximum question overlap with operand row and column labels; and fractions with overlapping question

years, known units, and compatible units. Row and column overlap use token-set Jaccard similarity. Unit compatibility checks typed categories and explicit scale mismatches without converting values.

These features express why two equally responsive answers need not carry the same evidence. A candidate supported by the queried year and table row differs from a coincidental numerical match elsewhere in the table. The three worldwide summaries contribute 24 dimensions. Four deterministic question cues—direct reading, ratio, change, and comparison—allow the predictor to condition on the requested operation. They are observable lexical indicators, not gold-program classes.

4.4 Persistence under interventions

Cross-world features distinguish a calculation that continues to fit the responses from unrelated matches that appear separately. Define

$$\mathcal{I} = \bigcap_w \mathcal{M}^w, \quad \mathcal{U} = \bigcup_w \mathcal{M}^w. \quad (8)$$

The persistence and switching summaries are

$$J = \frac{|\mathcal{I}|}{\max(1, |\mathcal{U}|)}, \quad (9)$$

$$S = \frac{1}{2} \sum_{(u,v) \in \mathcal{E}} \frac{|\mathcal{M}^u \triangle \mathcal{M}^v|}{\max(1, |\mathcal{M}^u \cup \mathcal{M}^v|)}. \quad (10)$$

Here $\mathcal{E} = \{(-, 0), (0, +)\}$ and $\triangle$ denotes symmetric difference. A soft joint-fit feature also retains information when no candidate meets every hard threshold:

$$F_{\text{joint}} = \max_{\pi \in \mathcal{P}} \frac{1}{1 + \max_w d_w(\pi)}, \quad (11)$$

with value zero for an empty candidate pool.

Together with existence, uniqueness, and log-count indicators, the joint block summarizes

whether common candidates involve the seed or another changed operand; agree with question-operation cues; overlap question years; and have compatible units and row/column labels. It has 16 dimensions. This construction asks more than whether all answers change in the expected direction: it asks whether one coordinate-identified computation continues to explain the trajectory.

4.5 A fixed error-risk learner

The final alignment vector and predictor are

$$A_{44} = \text{concat}(\psi_0, \psi_+, \psi_-, \chi, \omega), \quad (12)$$

$$z = \text{concat}(B, A_{44}) \in \mathbb{R}^{158}, \quad (13)$$

$$s = f_\theta(z), \quad (14)$$

where $\chi \in \mathbb{R}^{16}$ is the joint block and $\omega \in \mathbb{R}^4$ the question-cue block. All empty-set averages and maxima are defined as zero, with match-existence indicators retaining the missing-match distinction. Feature extraction reads no correctness labels, reference answers, or annotated solution programs. It needs no extra LLM call and enumerates $O(|\mathcal{O}||\mathcal{V}|^2)$ one-step candidates per triplet.

We fit the same supervised HGB configuration to each representation. For outer fold k , fitting uses only $\{(z_i, y_i) : \text{fold}(g_i) \neq k\}$; the held-out fold receives scores from that fitted predictor. The score ranks original-answer error risk. Standard tree optimization is unchanged: the contribution is the information supplied to the detector, not a new boosting algorithm.

5 Experimental Setup

Tracks and splits. The main tracks are Qwen3.5-4B and DeepSeek V4.1 Flash, with strict support reported in Table 1. Collection uses non-thinking profiles, temperature zero, and a 384-token output cap for both. We retain five saved source-group folds; each company/year report group stays within one test fold. A separate risk predictor is fitted per model track. This evaluates the representation across answer models, rather than transferring one fitted classifier without target-model labels. Appendix C reports additional Qwen tracks and an existing company-disjoint sensitivity analysis.

Comparators. G96 uses the original answer and evidence. The construction control adds the exact clean descriptors $[\mathbb{I}(e = 1), h, t]$. G96+Raw16

adds the bidirectional response block; Raw additionally supplies h, t (Equation 5). Curve29 retains the developed curve representation, including normalized slopes, asymmetry, direction, and confidence relations. PECR adds Arithmetic44 to Raw. The three final methods each consume three responses. We evaluate component removals from Arithmetic44 under the same configuration.

Training and reproducibility. All predictors use 250 HGB iterations, learning rate .05, at most 15 leaves per tree, minimum leaf size 20, L_2 regularization 1.0, and seed 20260928. No learner tuning is performed for the new attribution controls. A fresh environment reproduces all 60 released main/ablation fits exactly. The present control expansion performs 90 fixed fits and independently reproduces the Raw, Curve, and PECR OOF scores on both tracks. Versions, complete parameters, and positional feature schemas accompany the tables.

Metrics and uncertainty. We report error-positive AUROC and average precision (AP). The primary comparison is PECR minus Raw; Curve is a second equal-call comparator. Paired bootstrap draws resample entire source groups with the two methods' OOF scores kept together. Main and ablation analyses use 5,000 draws; the new attribution controls use 10,000. We retain each completed analysis's multiplicity family, stated with its table, rather than select the narrowest available interval. These are development evaluations on previously explored source questions. Intervals condition on saved OOF predictions and do not remove earlier method-selection effects. The distinct scope of company-held-out fitting and company-cluster resampling is documented in Appendix C.

6 Results and Analysis

6.1 Arithmetic alignment improves equal-budget diagnosis

PECR improves AUROC from .8035 to .8294 on Qwen and from .7818 to .8068 on DeepSeek (Table 2). The gains over Raw are +.0259 and +.0250, with adjusted paired intervals [.0060,.0453] and [.0125,.0378], respectively (Table 3). AP improves from .9266 to .9399 and from .8131 to .8297. Because the comparators share responses, cohorts, original-side features, edit descriptors, and learner settings, these differences isolate the

Table 2: Error ranking on the two generation-aligned response tracks. All methods use the same strict items, five source-group folds, and fixed HGB within each track.

Input	Dim.	Calls	Qwen3.5-4B		DeepSeek V4.1 Flash	
			AUROC	AP	AUROC	AP
Original-answer G96	96	1	0.7379	0.9003	0.7260	0.7490
G96 + construction metadata	99	1	0.7421	0.9033	0.7386	0.7622
G96 + Raw16	112	3	0.8023	0.9268	0.7819	0.8140
Raw: G96 + Raw16 + (h, t)	114	3	0.8035	0.9266	0.7818	0.8131
G96 + Curve29	125	3	0.8029	0.9296	0.7839	0.8184
PECR: Raw + Arithmetic44	158	3	0.8294	0.9399	0.8068	0.8297

Qwen: 1,597 items, 413 groups, 1,258 errors; DeepSeek: 1,537 items, 407 groups, 837 errors. Calls count the response slots consumed by a detector. Construction and its expected direction are supplied benchmark information. AP uses error as the positive class.

Table 3: Paired gains of PECR over equal-call comparators. Differences subtract the comparator from PECR.

Track	Comparator	Δ AUROC	Adjusted 95% CI	Δ AP	Adjusted 95% CI
Qwen3.5-4B	Raw	+0.0259	[+0.0060, +0.0453]	+0.0133	[+0.0034, +0.0246]
Qwen3.5-4B	Curve29	+0.0265	[+0.0067, +0.0456]	+0.0103	[+0.0017, +0.0197]
DeepSeek V4.1 Flash	Raw	+0.0250	[+0.0125, +0.0378]	+0.0166	[+0.0037, +0.0302]
DeepSeek V4.1 Flash	Curve29	+0.0229	[+0.0087, +0.0378]	+0.0113	[-0.0034, +0.0267]

Intervals retain the saved analyses: 5,000 paired source-group draws; six-contrast campaign correction for Qwen and three-contrast correction for DeepSeek, separately by metric. They condition on saved OOF predictions.

contribution of the added representation under the evaluated protocol.

PECR also improves AUROC over Curve29 on both tracks, by .0265 and .0229. Curve itself is close to Raw: slightly lower AUROC on Qwen and slightly higher on DeepSeek. The main result therefore favors connecting responses to arithmetic evidence over further describing the scalar trajectory alone. AP versus Curve improves on both point estimates, although its adjusted DeepSeek interval includes zero. We retain this distinction between ranking metrics rather than treat a positive AUROC interval as evidence for every metric.

6.2 The signal is not explained by edit magnitude alone

The new controls answer two attribution questions. First, edited responses add information beyond the original answer: G96+Raw16 improves AUROC over G96 by .0644 on Qwen and .0559 on DeepSeek, with adjusted intervals above zero. Second, this benefit is not reproduced by supplying only the clean construction descriptors. Raw exceeds the G96+construction model by .0614 and .0432, again with positive adjusted intervals (Table 4). These results support using the observed response dynamics rather than construction metadata alone.

Adding h, t to G96+Raw16 changes AUROC by only +.0012 on Qwen and -.0001 on DeepSeek, with both intervals spanning zero. Nevertheless, Raw retains these inputs, giving the baseline the same edit-scale information as the proposed method. PECR’s additional gain thus cannot be attributed merely to supplying h, t . The control covers the construction descriptors used by the clean comparator; it does not establish independence from every aspect of program-conditioned construction.

6.3 Different models benefit from different alignment blocks

Removing the 24 worldwide features reduces Qwen AUROC from .8294 to .8117, a full-minus-removal difference of .0177 with adjusted interval [.0033, .0326]. DeepSeek instead shows its clearest loss when the 16 cross-world features are removed: AUROC falls from .8068 to .7928, with difference .0140 [.0050, .0239]. Per-world evidence correspondence and persistence across interventions therefore make distinct conditional contributions (Table 5).

No block is uniformly essential across tracks. Qwen’s AUROC rises slightly when the joint or question-cue block is omitted; both adjusted intervals include zero. The question-cue block has no established incremental gain on either track.

Table 4: Attribution controls on the same strict cohorts. The construction model is $G96 + \mathbb{I}(e = 1), h, t]$ and consumes no edited-world responses.

Contrast	Qwen3.5-4B		DeepSeek V4.1 Flash	
	Δ AUROC	Adjusted 95% CI	Δ AUROC	Adjusted 95% CI
Construction model – G96	+0.0042	[-0.0105, +0.0198]	+0.0126	[-0.0027, +0.0287]
(G96 + Raw16) – G96	+0.0644	[+0.0264, +0.1007]	+0.0559	[+0.0269, +0.0855]
Raw – (G96 + Raw16)	+0.0012	[-0.0081, +0.0110]	-0.0001	[-0.0105, +0.0106]
Raw – construction model	+0.0614	[+0.0236, +0.0995]	+0.0432	[+0.0158, +0.0710]

Fixed post-hoc controls: 10,000 paired source-group bootstrap draws and Bonferroni correction over these four contrasts in both tracks (eight comparisons). AUROC and AP form separate families. G96 includes the original answer; this control isolates added construction descriptors, not all possible construction information.

Table 5: Fixed component removals from Arithmetic44 (AUROC). Differences are full PECR minus the ablated variant.

Variant	Dim.	Qwen3.5-4B		DeepSeek V4.1 Flash	
		AUROC	Δ [adjusted 95% CI]	AUROC	Δ [adjusted 95% CI]
Full PECR	158	0.8294	—	0.8068	—
w/o worldwide (24)	134	0.8117	+0.0177 [+0.0033, +0.0326]	0.8028	+0.0040 [-0.0072, +0.0155]
w/o cross-world (16)	142	0.8334	-0.0040 [-0.0138, +0.0059]	0.7928	+0.0140 [+0.0050, +0.0239]
w/o question cues (4)	154	0.8340	-0.0046 [-0.0119, +0.0028]	0.8016	+0.0052 [-0.0025, +0.0131]

All variants retain the same Raw block. Intervals use 5,000 paired source-group draws and Bonferroni correction across three block removals in two tracks (six comparisons per metric). A positive difference favors retaining the block.

These fixed removals support model-dependent contributions from alignment features, rather than an additive benefit from every block. Appendix Table 11 reports the AP ablations.

6.4 Why response direction is not enough

Direction violations are strongly associated with original-answer errors, but successful directional behavior does not resolve the task. Among answers aligned with the expected direction on both sides, 71.6% are still errors on Qwen and 49.0% on DeepSeek (Appendix Table 8). These rates are lower than each track’s overall error prevalence, so direction is informative, yet far from decisive. Flat and reversed responses describe other high-risk patterns. Arithmetic alignment supplies an additional view of these behaviors: which calculations fit, how many alternatives exist, and whether the same candidate remains compatible across worlds.

6.5 Robustness and remaining variation

An existing Qwen company-disjoint check holds out all years of a company together and averages metrics over five fixed five-fold partitions. PECR reaches mean AUROC .8199 versus .7901 for Raw, with paired gain .0299 and adjusted company-bootstrap interval [.0135, .0471]. This supports the main finding under stricter split granularity (Appendix C). Additional Qwen3.7-Flash

and Qwen3.8-Flash analyses give smaller gains and do not establish superiority over both comparators under their corrected criteria. We retain them alongside the two main tracks: the benefit is reproducible in the main evaluations, while its magnitude and statistical resolution vary with the response population.

7 Conclusion

Controlled evidence changes provide additional diagnostic evidence beyond an original answer, but response shape alone is an incomplete diagnostic. PECR connects a three-world trajectory to arithmetic correspondences in the underlying evidence and learns which combinations predict original-answer errors. Equal-budget gains on Qwen and DeepSeek, construction-information controls, and complementary ablation findings support this direction. The accompanying benchmark binds reviewed interventions, saved generations, labels, and reproducible evaluation, enabling both offline method comparison and new-model studies. Together, they make the relationship between evidence use and answer reliability a concrete, inspectable object of evaluation.

Limitations

The benchmark is program-conditioned and restricted to numerical financial QA. Construction direction and seed location are supplied metadata. All retained constructions received one human validity judgment, but there is no independent second review, documented blinding, or item-level subcriterion agreement; dependencies outside the supplied excerpt can remain unresolved. Forty-four direction-pending items are excluded from strict fitting. Arithmetic candidates are limited to bounded one-step table calculations, and matching does not recover a model’s internal reasoning or certify semantic correctness.

The source population supported method development. Grouped OOF evaluation and within-campaign corrections control specific comparisons, not the entire development history; new source questions remain necessary for independent confirmation. The two main tracks share questions, and a detector is trained on each track’s own labels. Additional model results are included in the appendix to show variation. We have not run all sampling-based uncertainty baselines at an equivalent collection budget.

Performance is conditional on strict parsing and label coverage. Labels follow a frozen numerical policy without unit conversion or percentage rescaling. Runtime identifiers do not independently certify provider checkpoints, and input reconstruction does not guarantee future output replay. We evaluate ranking rather than calibrated probabilities, autonomous financial decisions, or realized reviewer time savings.

Ethics and Reproducibility

Error-risk scores are intended to prioritize verification. They should not replace verification in financial decisions. The prepared artifact includes numerical features, separate labels, OOF scores, edit patches, provenance hashes, and evaluation code; it excludes credentials, model weights, raw reasoning, and complete financial passages. Rebuilding requests requires the exact FinQA source and retains its license notice. Source hashes, fixed evaluation folds, and a complete coverage ledger support reproducibility without redistributing complete financial records.

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A Exact Feature and Implementation Specification

Original-side G96. G96 concatenates nine fixed feature families: section-wise number counts (5), log statistics and type rates (28), input structure and response matching (26), typed number counts (12), numeric-cell positions and occupancy (5), pair-comparison summaries (4), lexical indicators (10), missing-region flags (3), and unit flags (3). The sum is 96. All fields are derived from the original question/evidence and the same model’s saved original answer. G96 is a handcrafted numerical and lexical summary, not a graph neural encoder. A historical duplicated feature name remains distinguished by its column index.

Raw16 and edit descriptors. For $\varepsilon = 10^{-9}$, the raw block is

$$R_{16} = [v^+, v^-, |v^+|, |v^-|, ev^+, ev^-, \mathbb{I}(ev^+ > 0), \mathbb{I}(ev^- > 0), \mathbb{I}(v^+ = 0), \mathbb{I}(v^- = 0), B_v, c^+ - c^0, c^- - c^0, U_{eq}, U_{present}, \mathbb{I}(e = 1)], \quad (15)$$

$$B_v = \frac{\min(|v^+|, |v^-|)}{\max(|v^+|, |v^-|) + \varepsilon}. \quad (16)$$

The two unit flags indicate identical normalized unit strings and three nonempty unit strings, respectively. The first does not imply the second. The separate descriptors are the signed actual seed edit $h = b^+ - b^0$ and $t = h/(|b^0| + \varepsilon)$. Primary concatenation order is $[G, R_{16}, h, t]$.

Curve comparator. The saved Curve33 archive retains the historical positional schema. The clean comparator removes positions 16, 24, 25, and 30 (zero-based): the inherited construction scalar, normalized second difference, slope difference, and magnitude-asymmetry feature. The remaining 29 columns include Raw16, h , t , normalized slopes and magnitudes, signed slope asymmetry, confidence-direction relations, confidence stability, signed response agreement, and a both-nonflat flag. G96+Curve29 has 125 input dimensions. It is the clean counterpart of the previously developed Curve30-A view, not the full original Curve33.

Historical Raw17. Raw17 appends a saved scalar μ named `construction_absolute_delta`. Its relation to the repaired construction is unresolved,

so the main views exclude it on both sides of every comparison. Table 6 supplies the requested Raw17 progression as a compatibility diagnostic. Reintroducing μ changes some point estimates; those values are not substituted into the clean primary results.

Candidate semantics. Sums use unordered distinct cell pairs; differences and ratios use ordered distinct pairs. When all reported units contain a percent marker, ratio becomes $100x/y$ and percentage change $100(x/y - 1)$ is added. Otherwise ratio is x/y . A denominator must be nonzero in every world. Units are inferred from cell text, row/column labels, and the table corner cell; known conflicting thousand/million/billion scales make a candidate unit-incompatible. Values are never converted. Years are recognized in question and operand row/column text. For a common candidate, known-unit and compatible-unit summaries require the corresponding condition in all worlds.

Question-operation agreement uses fixed lexical cues: ratio cues for ratios, change cues for percentage change, change or comparison cues for differences, and sum cues for addition. No gold-derived operation category is used. The extractor’s confidence field is allowlisted for response identity but is not read by Arithmetic44. Literal-answer/cell equality is recorded in diagnostic traces only and is not an additional feature in the 44-dimensional vector. These details prevent interpretation of the vector as a hidden solution-program verifier.

B Construction, Review, and Coverage

The initial model pass uses the recorded GPT-6.1-Sol service; unresolved drafts receive a GPT-6-Astra pass. The final 1,699 complete constructions comprise 1,191 completed at the first stage and 508 completed at the second. Model identifiers record construction provenance rather than independently verified checkpoints. Construction prompts require full positive/negative worlds, source-exact patches, dependency updates, and explicit unresolved status. They prohibit reference-answer, correctness-label, target-response, and OOF access. Seed cells and expected directions originate in the earlier program-conditioned construction stage; the repair models do not receive the full annotated program.

The supplied review records all 1,699 complete constructions as Valid. Reviewer iden-

Table 6: Historical-descriptor compatibility controls, refitted on the current cohorts. These are excluded from the main clean comparison.

Input	Dim.	Qwen		DeepSeek	
		AUROC	AP	AUROC	AP
G96	96	0.7379	0.9003	0.7260	0.7490
G96 + $[\mathbb{I}(e = 1), \mu, h, t]$	100	0.7505	0.9029	0.7536	0.7734
G96 + historical Raw17	113	0.8128	0.9311	0.7814	0.8139
G96 + historical Raw17 + (h, t)	115	0.8142	0.9306	0.7824	0.8090

μ denotes the inherited Raw17[16] scalar named construction_absolute_delta. Its alignment with repaired edits is unresolved; it is neither substituted by $|h|$ nor used by the proposed primary method.

Table 7: Arithmetic44 in fixed positional order. Fractions average over the relevant match set. Empty-set fractions, maxima, and averages equal zero.

Block	Dim.	Ordered components
Worldwise ψ_w	3×8	For $w = 0, +, -$: log-one-plus match count; unique match; seed-operand fraction; maximum row/question Jaccard; maximum column/question Jaccard; question-year overlap fraction; known-unit fraction; compatible-unit fraction.
Joint χ	16	All worlds have a match; common candidate exists; common candidate unique; log-one-plus common count; common/union Jaccard J ; switch fraction S ; common seed fraction; common changed-operand fraction; common question-operation agreement; common known-year fraction; common year-overlap fraction; common all-world known-unit fraction; common all-world compatible-unit fraction; maximum common row/question Jaccard; maximum common column/question Jaccard; best joint fit F_{joint} .
Question ω	4	Direct-read, ratio, change, comparison cues.

tity (reviewer1), one-reviewer count, and date (2026-10-06) were supplied separately by the project lead because the uploaded form’s identity/date cells were blank. These attributes are retained as reported provenance. Per-criterion verdicts were not recorded; no inter-rater agreement or blinding claim is made. The 44 direction-pending items remain deferred and outside both strict cohorts. The distributed package contains the aggregate review receipt; the original form and derived item-level receipt remain in the project audit archive.

Qwen’s 102 strict exclusions partition into 44 direction-pending, 22 feature/parse-ineligible, and 36 label-unknown items under the saved precedence rule. DeepSeek retains 1,537 of 1,597 attempts; all 60 excluded items remain in its coverage record. World-specific transport, schema, numeric parsing, and label statuses are preserved separately. In particular, an attempted item with an unknown label is not treated as an error, and a not-attempted candidate is not a parsing failure.

Generation alignment. The label source is the current original response for each track. G96 uses that same response; Raw and Arithmetic44 use its three-world triplet. DeepSeek G96 is native

to DeepSeek. Within-track comparisons preserve ID order, group assignments, labels, feature order, and fold assignments. The earlier generation-mismatched and shared-Qwen-feature results remain historical records and are not combined with the current main table.

Request reconstruction. The compact edit specification contains table-cell and text-span patches relative to the original FinQA evidence. The rebuild script checks the source hash, applies patches, renders the saved prompt, and verifies canonical request hashes. It decodes only the question, table, pre-text, and post-text needed for the request. Full financial evidence and instantiated prompts are obtained locally from the user’s source file rather than redistributed. The exact 9,888 hash matches establish input reconstruction, not replay of a mutable model endpoint’s outputs.

C Evaluation Details and Additional Controls

The reference runtime is Python 3.13.13, NumPy 2.5.3, SciPy 1.18.1, scikit-learn 1.9.0, and threadpoolctl 3.6.0. Each HGB fit uses one CPU thread. At these sample sizes the saved automatic early-stopping setting does not activate, and all fitted

Table 8: Response patterns and original-answer errors.

Pattern	Qwen N (error %)	DeepSeek N (error %)
Both sides aligned	1,081 (71.6)	1,319 (49.0)
Mixed directions	316 (92.1)	70 (78.6)
Both sides reversed	109 (96.3)	94 (89.4)
One side flat	43 (97.7)	18 (100.0)
Both sides flat	48 (95.8)	36 (94.4)

All strict items are retained. Categories are mutually exclusive; flatness is exact zero change. Alignment uses the saved expected direction.

Table 9: Full construction-frame accounting. The last three rows partition the 2,225 requested candidates per track; unrequested records remain separate.

Partition	Qwen	DeepSeek
Requested construction frame	2,225	2,225
Unrequested records	16	16
Requested, not attempted	526	628
Attempted, strict	1,597	1,537
Attempted, excluded	102	60

models complete 250 iterations. Reference OOF equality is checked numerically, not inferred from rounded metrics. The new controls retain source hashes and a protocol fixed before their 90 fits.

AP differences for the construction-information controls appear in Table 10. AP comparisons are within track because positive-class prevalence differs: 78.8% for Qwen and 54.5% for DeepSeek. A high AP on Qwen must be interpreted against that high error prevalence.

Multiplicity families. The saved Qwen main campaign corrects six primary comparisons, while DeepSeek corrects three; each uses 5,000 paired source-group draws with seed 20260928. The ablation corrects three removals across two tracks (six AUROC contrasts, with AP handled separately). The new construction controls correct four contrasts across two tracks (eight per metric), using 10,000 draws and the same bootstrap seed. Percentile bounds use $\alpha/(2K)$ and $1 - \alpha/(2K)$ for family size K and $\alpha = .05$. These within-analysis corrections do not represent adjustment for all earlier research attempts.

Company-disjoint fitting. The Qwen sensitivity analysis uses all 1,597 items, 413 reports, and 117 companies. Five predetermined partition seeds assign entire companies to five folds, with all five runs reported. The summary averages complete-OOF metrics across runs; it neither picks the best partition nor treats repeated

predictions as independent samples. Mean AUROC is .7901 (Raw), .7995 (Curve29), and .8199 (PECR); mean AP is .9235, .9290, and .9362. Company-paired bootstrap resampling averages the five run-level metrics inside each draw. The PECR gains are .0299 [.0135,.0471] over Raw and .0205 [.0030,.0375] over Curve under the original four-comparison correction. This is stricter split granularity on the existing development population, not new-source confirmation.

Additional model evidence. Qwen3.7-Flash uses 1,129 items in 371 report groups and 113 companies; Qwen3.8-Flash uses 1,430 items in 401 groups and 116 companies. Both use generation-aligned native labels and features with the same clean dimensions (114, 125, 158). The smaller Raw-relative gains, .0080 and .0135, are retained in Table 12. Their corresponding Curve AUROCs are .8312 and .8336. The first track does not establish a Raw-relative gain under its corrected interval; the second does, but not a gain over Curve. Coverage and response populations differ, so these differences do not isolate model architecture as the cause.

Relationship to earlier experiments. Earlier manuscript versions centered on Curve33, historical construction semantics, and different original-response targets. Their headline scores, old Curve-family ablations, MultiHiertt boundary checks, and neural explorations do not evaluate the current Raw+Arithmetic44 method on this release. They remain archived development evidence rather than being inserted into the new main comparison. The current manuscript reports the clean Raw/Curve controls, repaired-construction cohorts, and Arithmetic44 component removals consistently.

Table 10: AP attribution controls with the same eight-comparison correction as Table 4.

Contrast	Qwen		DeepSeek	
	Δ AP	Adjusted 95% CI	Δ AP	Adjusted 95% CI
Construction model – G96	+0.0030	[-0.0053, +0.0114]	+0.0132	[-0.0062, +0.0333]
(G96 + Raw16) – G96	+0.0264	[+0.0036, +0.0473]	+0.0650	[+0.0337, +0.0979]
Raw – (G96 + Raw16)	-0.0002	[-0.0044, +0.0045]	-0.0009	[-0.0129, +0.0120]
Raw – construction model	+0.0233	[+0.0012, +0.0450]	+0.0509	[+0.0207, +0.0805]

Table 11: Fixed component removals from Arithmetic44 (AP). Differences are full PECR minus the ablated variant.

Variant	Dim.	Qwen3.5-4B		DeepSeek V4.1 Flash	
		AP	Δ [adjusted 95% CI]	AP	Δ [adjusted 95% CI]
Full PECR	158	0.9399	—	0.8297	—
w/o worldwide (24)	134	0.9319	+0.0080 [-0.0005,+0.0182]	0.8326	-0.0029 [-0.0144,+0.0085]
w/o cross-world (16)	142	0.9420	-0.0021 [-0.0063,+0.0024]	0.8185	+0.0112 [+0.0021,+0.0216]
w/o question cues (4)	154	0.9425	-0.0026 [-0.0060,+0.0006]	0.8285	+0.0012 [-0.0068,+0.0097]

All variants retain the same Raw block. Intervals use 5,000 paired source-group draws and Bonferroni correction across three block removals in two tracks (six comparisons per metric). A positive difference favors retaining the block.

Table 12: Additional saved Qwen response tracks with the clean feature definitions. Each track refits its own detector on its own labels.

Track	N	Raw AUROC	PECR AUROC	PECR – Raw CI	PECR – Curve CI
Qwen3.7-Flash	1,129	.8347	.8428	[-.0039,.0200]	[-.0003,.0226]
Qwen3.8-Flash	1,430	.8285	.8420	[.0006,.0260]	[-.0033,.0212]

Intervals retain the saved company-cluster analyses: four-comparison correction for Qwen3.7 (joint with the company-disjoint analysis), two-comparison correction for Qwen3.8. Their prediction folds are source-group folds, not company-disjoint folds. Neither track meets its corrected criterion of outperforming both comparators.